15.2 Kinetic Theory

Question Paper

Course	CIE A Level Physics
Section	15. Ideal Gases
Topic	15.2 Kinetic Theory
Difficulty	Hard

Time allowed: 40

Score: /28

Percentage: /100

Question 1a

A sealed container A has the shape of a rectangular prism and contains an ideal gas. The dimensions of the container are x, y and z as shown in Fig. 1.1.

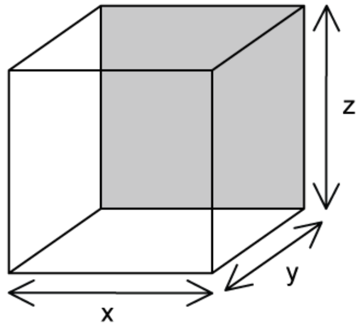

Fig. 1.1

- The average force exerted by the gas on the bottom wall of the container is F
- There are n moles of gas in the container
- The temperature of the gas is T

Obtain an expression for the height of the container in terms of F, n and T.

[2 marks]

Question 1b

A second container B contains the same ideal gas.

The following information is known:

- The pressure in B is a quarter of the pressure in A
- The volume of B is five times the volume of A
- Container B contains half the number of molecules as A
- The temperature of container B is 800 K

Calculate the average translational kinetic energy of the molecules in the gas of container A.

[3 marks]

Question 1c

Calculate the value of the ratio of the r.m.s. speed of the particles in containers A and B.

[2 marks]

Question 2a

The kinetic theory of gases assumes that collisions between particles and between particles and the walls of their container are perfectly elastic.

(i)
Describe what is meant by a *perfectly elastic collision*

[1]

(ii)
Describe the nature of the forces between the molecules

[1]

(iii)
Describe, with a reason, the type of speed of the molecules considered in the kinetic theory.

[2]

[4 marks]

Question 2b

A gas collimator is used to measure the average velocity of particles in a gas.

The collimator in Fig. 1.1 controls the pressure and the volume and it maintains a gas at a constant temperature.

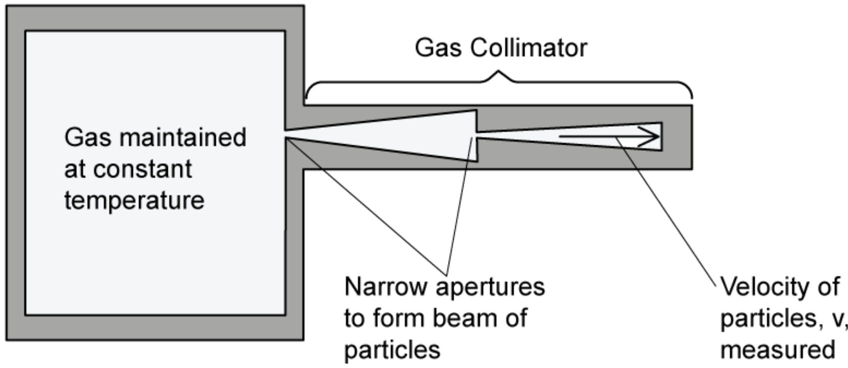

Fig. 1.1

Explain why two narrow apertures are required in the collimator to measure the velocity of the particles inside.

[3 marks]

Question 2c

A paint manufacturer is measuring the average velocity of a sample of propane particles used to make spray paint. At a temperature of 9 °C the propane molecules have a velocity of 399.00 m s^{-1} with an uncertainty of 0.5 %.

Propane has a molecular mass of 44.097 g mol^{-1}.

Determine whether the propane sample behaves as an ideal gas.

[6 marks]

Question 3a

A sealed box contains 3.0 moles of a monotonic ideal gas at a pressure of 6.1×10^5 Pa. The total internal energy of the gas is 7.5 kJ.

Calculate the volume of the gas.

[3 marks]

Question 3b

The molar mass of the gas in the box is 5.9 g mol^{-1}. Assume that there are perfectly elastic collisions between the particles and the box walls and that the particles travel perpendicular to the walls.

Calculate the magnitude of the impulse when a gas atom collides with the walls of the box.

[3 marks]

Question 3c

The box containing the gas is opened. Additional molecules enter the box. The number of gas atoms is increased by y%.

The box is then resealed and the temperature of the remaining gas atoms increased by 102.5 °C. The new pressure in the box is 2% of the original pressure.

Find the value of y.

[2 marks]

Model Answers: Hard

1a

(a) Obtain an expression in terms of *F, n* and *T* for the height of the container:

- Surface pressure: $P = \frac{F}{A}$
- Ideal gas equation: $PV = nRT$
- Combine the equations: $\left(\frac{F}{A}\right)V = nRT$ [1 mark]

Substitute and rearrange the dimensions *l*, *h*, and *w* into the equation:

- Area: $A = xy$
- Volume: $V = xyz$
- Substitute: $\frac{F \times xyz}{xy} = nRT \quad \Rightarrow \quad Fz = nRT$
- Rearrange for the height: $z = \frac{nRT}{F}$ [1 mark]

[Total: 2 marks]

1b

(b) Calculate the average translational kinetic energy of the molecules in the gas of container A:

List the known quantities:

- Pressure of gas in B: $P_B = \frac{1}{4}P_A$

- Volume of B: $V_B = 5V_A$
- Number of molecules in B: $n_B = \frac{1}{2}n_A$
- Temperature of gas in B: $T_B = 800$ K
- Molar gas constant, $R = 8.31$ J K^{-1} mol^{-1}
- Boltzmann constant, $k = 1.38 \times 10^{-23}$ J K^{-1}

Write expressions in terms of the ideal gas equation for the gas contained in A and B:

- Gas in container A: $P_A V_A = n_A R T_A$
- Gas in container B: $P_B V_B = n_B R T_B \quad \Rightarrow \quad \left(\frac{P_A}{4}\right) \times (5V_A) = \left(\frac{n_A}{2}\right) R \times 800$ [1 mark]
- Simplify: $\frac{5}{4} P_A V_A = 400 n_A R \quad \Rightarrow \quad P_A V_A = 320 n_A R$

Equate the equations in terms of R to find the temperature in container A:

- Gas in container A: $P_A V_A = n_A R T_A \quad \Rightarrow \quad R = \frac{P_A V_A}{n_A T_A}$
- Gas in container B: $P_A V_A = 320 n_A R \quad \Rightarrow \quad R = \frac{P_A V_A}{320 n_A}$
- Equate & simplify: $\frac{P_A V_A}{n_A T_A} = \frac{P_A V_A}{320 n_A}$
- Temperature of gas A: $T_A = 320$ K [1 mark]

Calculate the average translational kinetic energy of the molecules in the gas of container A:

- Average kinetic energy: $E_k = \frac{3}{2} kT$

- $E_k = \frac{3}{2} \times (1.38 \times 10^{-23}) \times 320$
- $E_k = 6.624 \times 10^{-21}$ J [1 mark]

[Total: 3 marks]

1c

(c) Calculate the value of the ratio of the r.m.s. speed of the particles in containers A and B:

List the known quantities:

- Average kinetic energy of molecules in A: $E_k = 6.624 \times 10^{-21}$ J (from **(b)**)
- Temperature of gas in container B, $T_B = 800$ K
- Mass of molecules in container B: $m_B = \frac{m_A}{2}$

Obtain expressions for the r.m.s. speed of the molecules in containers A and B:

- R.m.s. speed: $\frac{1}{2} m \langle c^2 \rangle$
- Container A: $\frac{1}{2} m \langle {c_A}^2 \rangle = 6.624 \times 10^{-21}$ J
- Container A: $c_A = \sqrt{\frac{(2 \times 6.624 \times 10^{-21})}{m}}$
- Container B: $\frac{1}{2}\left(\frac{m}{2}\right)\langle {c_B}^2 \rangle = \frac{3}{2} \times (1.38 \times 10^{-23}) \times 800$
- Container B: $c_B = \sqrt{\frac{(4 \times 1.656 \times 10^{-20})}{m}}$ [1 mark]

Obtain a ratio for r.m.s. speed of molecules in containers A and B:

- Ratio: $\frac{c_A}{c_B} = \frac{\left(\sqrt{\frac{1.32 \times 10^{-20}}{m}}\right)}{\left(\sqrt{\frac{6.624 \times 10^{-20}}{m}}\right)}$
- Ratio: $\frac{c_A}{c_B} = \sqrt{\frac{1.32 \times 10^{-20}}{6.624 \times 10^{-20}}}$
- Ratio: $\frac{c_A}{c_B} = 0.199 = 0.2$ [1 mark]

[Total: 2 marks]

The **mass of the molecules** in container B is **half** that in container A. This is because the mass of the molecules comes from the number of molecules in the container. If the number of molecules is half, then the mass is half.

2a

(a)

(i) A perfectly elastic collision is when:

Any **one** from:

- Kinetic energy is conserved during the collision; [1 mark]
- There is no energy transferred to heat / potential / sound energy **OR** wasted; [1 mark]

(ii) The forces between the molecules are:

- Not forces of attraction or repulsion; [1 mark]

(iii) The speed of the molecules considered in the kinetic theory is:

- The average speed; [1 mark]

This is because:

- The number of molecules of gas in a container is very large; [1 mark]

[Total: 4 marks]

2b

(b) Two narrow apertures are required in the collimator to measure the velocity of the particles inside because:

- Particles move in a constant random motion (Brownian motion); [1 mark]

Explain why the two narrow apertures are required:

- Only particles moving perpendicular to the direction of the beam / vertically pass through to be measured at the end; [1 mark]
- The particles are not travelling straight through the detector, so there is no error

- Only particles moving perpendicular to the direction of the beam / vertically pass through to be measured at the end; [1 mark]
- The particles are not travelling straight through the detector, so there is no error in the measured velocity; [1 mark]

[Total: 3 marks]

Understanding how a collimator works is **not** part of your **exam** syllabus **BUT** hard questions will ask you to **apply** your **knowledge** from the syllabus to a new situation.

2c

(c) Determine whether the propane sample behaves as an ideal gas:

- Molar mass of propane, M_r = 44.097 g mol^{-1} = 44.097 × 10^{-3} kg mol^{-1}
- Average velocity of particles, c = 399.00 m s^{-1}
- Uncertainty in velocity = 0.5 %
- Temperature, T = 9 °C = 9 + 273.15 = 282.15 K
- Avogadro constant, N_A = 6.02 × 10^{23} mol^{-1}
- Boltzmann constant, k = 1.38 × 10^{-23} J K^{-1}

Obtain an expression for the molar mass in terms of Avogadro's constant and calculate the mass of one propane molecule, m:

- $n = \frac{N}{N_A}$ and $n = \frac{m}{M_r}$
- $\frac{N}{N_A} = \frac{m}{M_r}$ and $m = \frac{NM_r}{N_A}$ where N = 1
- So, $m = \frac{M_r}{N_A}$ [1 mark]
- $m = \frac{44.097 \times 10^{-3}}{6.02 \times 10^{23}}$
- $m = 7.33 \times 10^{-26}$ kg [1 mark]

Use the average translational kinetic energy to calculate the average velocity, c:

- $E_k = \frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$
- $c = \sqrt{\frac{3kT}{m}}$ [1 mark]
- $c = \sqrt{\frac{3 \times (1.38 \times 10^{-23}) \times 282.15}{7.33 \times 10^{-26}}}$
- $c = 399.198 \text{ m s}^{-1}$ [1 mark]

Determine whether the propane behaves like an ideal gas:

- $\% \, uncertainty = \frac{calculated \ velocity - known \ velocity}{known \ velocity} \times 100$
- $\% \, uncertainty = \frac{399.20 - 399}{399} \times 100$
- $\% \, uncertainty = 0.05\%$ [1 mark]
- Hence, the velocity obtained is accurate to less than 0.5% so it does behave like an ideal gas; [1 mark]

[Total: 6 marks]

The velocity in the question is given accurately to **2 d.p**. so it is good practice to use that **same accuracy** when calculating your uncertainty.

3a

(a) Calculate the volume of the gas:

List the known quantities:

- Number of moles, $n = 3.0$
- Pressure, $p = 6.1 \times 10^5$ Pa
- Total internal energy, $U = 7.5$ kJ $= 7.5 \times 10^3$ J
- Avogadro constant, $N_A = 6.02 \times 10^{23}$ mol^{-1}
- Boltzmann constant, $k = 1.38 \times 10^{-23}$ J K^{-1}

- Molar gas constant, $R = 8.31$ J K^{-1} mol^{-1}

Calculate the average energy of the atoms in the gas, E_k:

- $E_k = \frac{U}{N} = \frac{U}{nN_A}$

- $E_k = \frac{7.5 \times 10^3}{3 \times (6.02 \times 10^{23})} = 4.15 \times 10^{-21}$ J [1 mark]

Calculate the temperature of the gas:

- Average kinetic energy: $E_k = \frac{3}{2}kT \quad \Rightarrow \quad T = \frac{2E_k}{3k}$

- $T = \frac{2 \times (4.15 \times 10^{-21}\ \text{J})}{3 \times (1.38 \times 10^{-23})} = 200.48$ K [1 mark]

Use the ideal gas equation to calculate volume, V:

- Ideal gas equation: $pV = nRT \quad \Rightarrow \quad V = \frac{nRT}{p}$

- $V = \frac{3 \times 8.31 \times 200.48}{6.1 \times 10^5} = 8.19\ \text{m}^3$ [1 mark]

[Total: 3 marks]

3b

(b) Calculate the magnitude of the impulse when a gas atom collides with the walls of the box:

List the known quantities:

- Number of moles, $n = 3.0$ (From part (a))
- Pressure, $p = 6.1 \times 10^5$ Pa (From part (a))
- Total internal energy, $U = 7.5$ kJ $= 7.5 \times 10^3$ J (From part (a))
- Average kinetic energy of atoms, $E_k = 4.15 \times 10^{-21}$ J (From part (a))
- Molar mass, $M_r = 5.9$ g mol^{-1} $= 5.9 \times 10^{-3}$ kg mol^{-1}
- Avogadro constant, $N_A = 6.02 \times 10^{23}$ mol^{-1}

Obtain an expression for the molar mass in terms of Avogadro's constant and calculate the mass of one propane molecule, *m*:

- $n = \frac{N}{N_A}$ and $n = \frac{m}{M_r}$
- $\frac{N}{N_A} = \frac{m}{M_r}$ and $m = \frac{NM_r}{N_A}$ where $N = 1$
- So, $m = \frac{M_r}{N_A}$ [1 mark]
- $m = \frac{5.9 \times 10^{-3}}{6.02 \times 10^{23}}$
- $m = 9.8 \times 10^{-27}$ kg [1 mark]

Use the average translational kinetic energy to calculate the average velocity, *c*:

- $E_k = \frac{1}{2}m\langle c^2 \rangle$
- $c = \sqrt{\frac{2E_k}{m}}$
- $c = \sqrt{\frac{2 \times (4.15 \times 10^{-21})}{9.8 \times 10^{-27}}}$
- $c = 920.29 \text{ m s}^{-1}$ [1 mark]

[Total: 3 marks]

3c

(d) Find the value of *y:*

List the known quantities:

- Original number of moles, $n = 3.0$ (from **(a)**)
- Original pressure, $p_1 = 6.1 \times 10^5$ Pa (from **(a)**)
- Volume of gas, $V = 8.19$ m^3 (from **(a)**)
- Original temperature, $T_1 = 200.48$ K (from **(a)**)
- New temperature, $T_2 = 200.48 + 102.5 = 302.98$ K
- New pressure, $p_2 = 0.02 \times p_1 = 0.02 \times (6.1 \times 10^5) = 1.22 \times 10^4$ Pa
- Molar gas constant, $R = 8.31$ J K^{-1} mol^{-1}

Calculate the new number of moles, n:

- Ideal gas equation: $pV = nRT \quad \Rightarrow \quad n = \frac{p_2 V}{R T_2}$

- $m = \dfrac{5.9 \times 10^{-3}}{6.02 \times 10^{23}}$
- $m = 9.8 \times 10^{-27}$ kg [1 mark]

Use the average translational kinetic energy to calculate the average velocity, *c*:

- $E_k = \dfrac{1}{2}m\langle c^2 \rangle$
- $c = \sqrt{\dfrac{2E_k}{m}}$
- $c = \sqrt{\dfrac{2 \times (4.15 \times 10^{-21})}{9.8 \times 10^{-27}}}$
- $c = 920.29$ m s^{-1} [1 mark]

[Total: 3 marks]

3c

(d) Find the value of *y:*

List the known quantities:

- Original number of moles, $n = 3.0$ (from **(a)**)
- Original pressure, $p_1 = 6.1 \times 10^5$ Pa (from **(a)**)
- Volume of gas, $V = 8.19$ m^3 (from **(a)**)
- Original temperature, $T_1 = 200.48$ K (from **(a)**)
- New temperature, $T_2 = 200.48 + 102.5 = 302.98$ K
- New pressure, $p_2 = 0.02 \times p_1 = 0.02 \times (6.1 \times 10^5) = 1.22 \times 10^4$ Pa
- Molar gas constant, $R = 8.31$ J K^{-1} mol^{-1}

Calculate the new number of moles, *n*:

- Ideal gas equation: $pV = nRT \quad \Rightarrow \quad n = \dfrac{p_2 V}{RT_2}$

- $n = \frac{(1.22 \times 10^4) \times 8.19}{8.31 \times 302.98} = 3.98 \text{ mol}$ [1 mark]

Calculate *y*:

- $y = \frac{difference\ in\ moles}{original\ number\ of\ moles} \times 100\%$
- $y = \frac{3.98 - 3}{3} \times 100 = 32.7\ \%$ [1 mark]

[Total: 2 marks]

Δ*T* can be in °C or K, as the temperature difference is the **same** for both scales, as it is